Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

	Section A		**3*10=30**
1)	Find the union of the given pair of simple graphs.	**3**	(CO2, L2)
2)	Find the union of the given pair of simple graphs.	**3**	(CO2, L2)
3) a)	Explain the Chinese postman problem.	**3**	(CO3, L5)
	OR		
b)	Define a spanning tree. Also, find a spanning tree of (i) (ii) . 2,3 5	**3**	(CO2, L3)
4)	What is meant by “influence graph”? Give example.	**3**	(CO3, L1)
5)	What is meant by “influence graph”? Give example.	**3**	(CO3, L1)
6)	Construct a pair of non-isomorphic graphs on six vertices which have the same number of edges and same degree sequence.	**3**	(CO2, L6)
7)	Describe a cubic graph. What is the number of vertices in a cubic graph with 15 edges?	**3**	(CO2, L3)
8) a)	Define a spanning tree. Also, find a spanning tree of (i) (ii) . 2,3 5	**3**	(CO2, L3)
	OR		
b)	Distinguish Eulerian and Hamiltonian graphs. Give an example of a graph that is neither Eulerian nor Hamiltonian.	**3**	(CO1, L4)
9)	How many leaves does a full 3 – ary tree with 100 vertices have?	**3**	(CO2, L4)
10)	Define a spanning tree. Find a spanning tree of (i) (ii) . 5 5	**3**	(CO1, L2)
	Section B		**7*3=21**
11)	Explain N−queens problem and use backtracking to solve − queens problem for = 4.	**7**	(CO3, L4)
12) a)	i) Prove that an undirected graph has even number of vertices of odd degree. ii) Draw the graph represented by the given adjacency matrix . 0 1 3 1 1 1 2 1 = [ ] 2 1 2 0 1 0 0 2	**7**	(CO2, L4)

OR

b)	Explain N−queens problem and use backtracking to solve − queens problem for = 4.	**7**	(CO3, L4)
13)	Find the number of paths of length four from the vertex to vertex in a simple graph G given by	**7**	(CO3, L5)

Section C **7*4=28**

14)	Explain Konigsberg bridge problem and Euler's contribution in solving it in detail.	**7**	(CO1, L2)
15)	(i) Define (a) -ary tree (b) full -ary tree. Prove that a tree with vertices have −1 edges. (ii) How many pendant vertices will a complete binary tree on vertices have? Justify your answer.	**7**	(CO2, L2)
16) a)	Use Huffman coding to encode the following symbols with frequencies listed below: : 0.20, : 0.10, : 0.15, : 0.25, : 0.30 What is the average number of bits used to encode a character? OR (i) Explain depth-first search algorithm. Use depth-first search to find a spanning tree of . 5 (ii) Is every internal vertex of a tree a cut-vertex of it? Justify your answer	**7**	(CO1, L3)

OR

b)	(i) Define adjacency matrix and incidence matrix. Represent and 5 with adjacency and incidence matrices. 2,3 (ii) List out any four properties of each of the adjacency and incidence matrices of a graph.	**7**	(CO3, L3)
17)	If G is a simple connected planar graph, then prove that there is a vertex of degree not exceeding five.	**7**	(CO1, L4)

Section D **7*3=21**

18)	Find the length of a shortest path between and in the given weighted graph by using Dijkstra's algorithm	**7**	(CO3, L4)
19) a)	Explain the isomorphism of graphs. Examine whether the following graphs are isomorphic. Justify.	**7**	(CO3, L5)

OR

b)	Use Huffman coding to encode the following symbols with frequencies listed below. : 0.10, : 0.25, : 0.05, : 0.15, : 0.30, : 0.07, : 0.08. What is the average number of bits used to encode a character?	**7**	(CO3, L4)
20)	Define universal address system. For a given ordered rooted tree , find the preorder traversal and postorder traversal.	**7**	(CO1, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use